



Tech. Degree III Semester Supplementary Examination May 2017

ME 15-1304 FLUID MECHANICS
(2015 Scheme)

Time: 3 Hours

Maximum Marks: 60

PART A

(Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) Explain the concept of normal and shear stresses over an elementary surface.
- (b) In a wide rectangular open channel, the velocity distribution in the vertical is given by $u = 1.25y^{1/6}$. Assuming depth of flow to be 2m, determine the discharge flowing per unit width of the channel.
- (c) A circular and square pipe are of equal cross sectional area. For the same flow rate, determine which section will lead to a higher value of friction factor.
- (d) Determine the optimum diameter of the pipe required to carry $0.1 \text{ m}^3/\text{s}$ of crude oil ($\rho = 950 \text{ kg/m}^3$, $\mu = 8 \times 10^{-3} \text{ kg/ms}$) and still maintaining laminar flow.
- (e) A pitot static tube is mounted on an aircraft travelling at a speed of 300 kmph against a wind velocity of 20 kmph. If the specific weight of air is 12 N/m^3 , determine the pressure difference the instrument will register.
- (f) If $u = -ay$ and $v = ax$ are the velocity components for two dimensional flow, determine the equation of stream line passing through (3, 2).
- (g) Find the circulation around the closed curve defined by $y=1, x=2, y=4, x=4$, when the flow field is given by $u = 16y - 8x, v = 8y - 7x$.
- (h) Explain the concept of circulation and vorticity.
- (i) If velocity distribution in laminar boundary layer over a flat plate is assumed as $u = ay + by^3$, determine the velocity distribution law using necessary boundary conditions.
- (j) Define boundary layer thickness.

PART B

(4 × 10 = 40)

- II. (a) In three communicating vessels, pistons are exposed to the forces F_1, F_2, F_3 as shown in figure 1. (3)
Given $F_1 = 1100 \text{ N}, F_2 = 600 \text{ N}, F_3 = 1000 \text{ N}, A_1 = 0.04 \text{ m}^2, A_2 = 0.02 \text{ m}^2, A_3 = 0.03 \text{ m}^2, \rho = 1000 \text{ kg/m}^3$. Take $g = 10 \text{ m/s}^2$. Determine the difference in heights Δh_1 and Δh_2 .

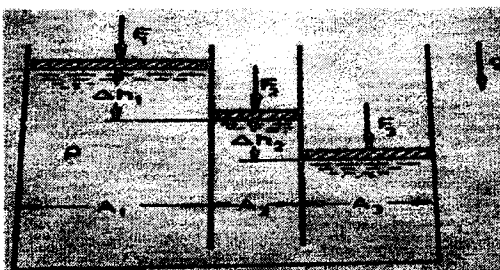


Figure 1

(P.T.O.)

- (b) A roller gate as shown in figure 2 has a span of 5m. Determine the magnitude and direction of the resultant force on the cylinder when water just begins to over flow. Neglect weight of the gate. (4)

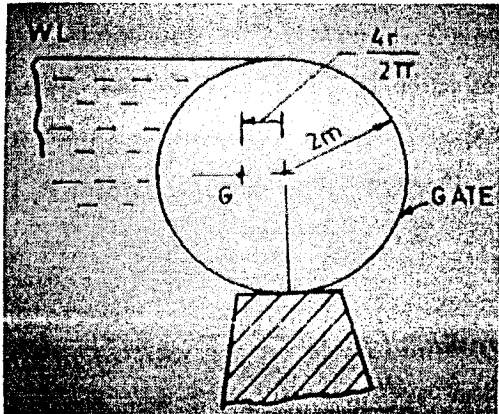


Figure 2

- (c) Determine the missing component u of the velocity distribution of an incompressible fluid. Given $v = ax^3 - by^2 + cz^2$ and $w = bx^3 - cy + axz^2$. (3)

OR

- III. (a) Water is filled partially in a cylinder of 1m diameter and rotated at 150 rpm. The cylinder is empty at the bottom surface up to a radius of 0.4 m as shown in figure 3. Determine the pressure at the extreme bottom edge. Also calculate the height of liquid at the edge. (3)

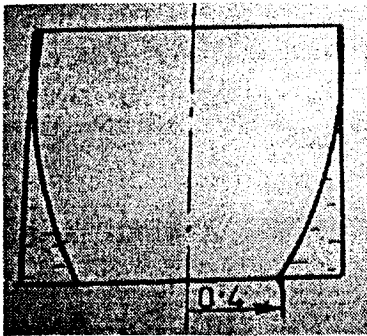


Figure 3

- (b) Two spheres, one heavier and weighing 12000 N and of diameter 1.2 m and the other lighter and weighing 4000 N, are tied with a rope and placed in water. It was found that the spheres floated vertically with the lighter sphere just submerging as shown in figure 4. Determine the diameter of the lighter sphere and the tension in the rope. (4)

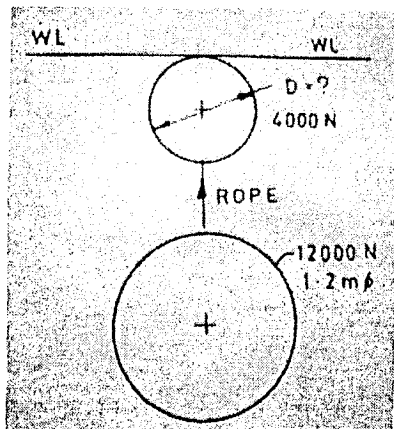


Figure 4

- (c) A pipe line is set up to draw water from a reservoir. The pipe line has to go over a barrier which is above the water level as shown in figure 5. The outlet is 8 m below water level. Determine the maximum height of the barrier, if the pressure head at this point should not fall below 1.0 m of water to avoid cavitation. Take atmospheric pressure head as 10.3 m. (3)

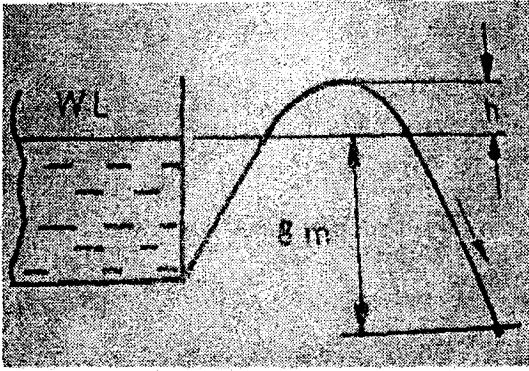


Figure 5

- IV. (a) The flow in a pipe of 100 mm diameter with Reynolds number value of 10^5 is found to have a friction factor $f = 0.032$. Determine the thickness of laminar sub layer. Also indicate whether the pipe is hydraulically smooth or not, if the roughness height is 0.4 mm. (5)
- (b) A pipe bend to 90° with moderate value of centerline bend radius (R_0) to pipe diameter (D) ratio, is used as a flow measuring device, by relating Q to pressure difference between outside (at R_2 radius) and inside (at R_1 radius) of the bend as shown in figure 6. Assume that the fluid flows at a constant velocity around the bend. Determine the discharge coefficient C_d of the measuring device. (5)

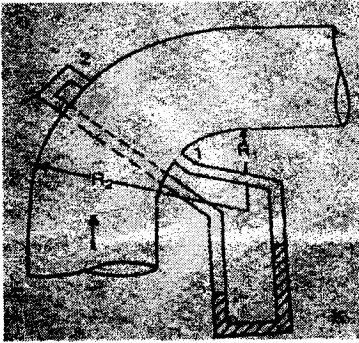


Figure 6

OR

- V. (a) Calculate the maximum power available at the far end of a hydraulic pipe line 5 km long and 200 mm in diameter when water at 10 MPa is fed in at the inlet end. Assume $f = 0.007$. (5)
- (b) Water flows vertically downward through a 15 cm diameter pipe with a velocity of 2.4 m/s. The pipe suddenly enlarges to 30 cm diameter. Find the loss of head and also the loss of head if the flow is reversed, the coefficient of contraction now being 0.62. (5)

- VI. (a) If the potential function $\phi = \frac{-A}{2\pi} \ln r$, where A is a positive constant, determine the stream function Ψ and plot typical equipotential lines and stream lines. Identify the flow pattern. (5)
- (b) Superposition of a circulation of strength $-\Gamma$, on uniform flow of velocity U , normal to a circular cylinder of radius ' a ', produces a streamline pattern represented by the stream function $\Psi = U \left(r - \frac{a^2}{r} \right) \sin \theta + \frac{\Gamma}{2\pi} \ln r$. Sketch neatly the flow patterns for different values of ' Γ ' clearly marking the positions of stagnation points. (5)

OR

(P.T.O.)

- VII. (a) In a two dimensional flow the x and y directional velocities u and v are given by (5)

$$u = -\frac{x}{x^2 + y^2}, \quad v = -\frac{y}{x^2 + y^2}. \quad \text{Show that the flow is steady and check whether the}$$

flow is irrotational. Also determine the acceleration components of the flow.

- (b) A source of strength $3.0 \text{ m}^3/\text{sec-m}$ is located at the origin and a two dimensional uniform rectilinear flow at 1.0 m/sec velocity in positive x -direction is superimposed on it. Determine (i) location of stagnation point (ii) maximum value of stream line passing through stagnation point (iii) velocity at $(-0.30, 1.5)$. (5)

- VIII. (a) What is the physical significance of displacement and momentum thicknesses of a boundary layer? (5)

- (b) Air flows at 30 m/s on a wide plate kept parallel to the air stream. Assuming the laminar boundary layer to follow the velocity distribution $\frac{u}{U} = \frac{y}{\delta}$, determine the rate of flow across the section 'bc', if section 'cd' is located 0.25 m downstream of the leading edge of the plate, where boundary layer thickness is 0.15 mm as shown in figure 7. (5)

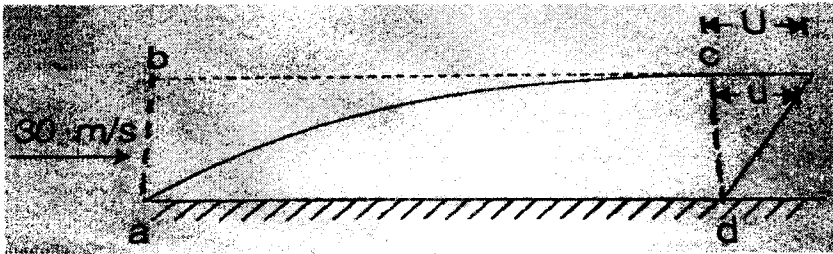


Figure 7

OR

- IX. (a) Briefly explain the boundary layer separation and its control. (4)

- (b) Air at 20°C ($\rho = 1.208 \text{ kg/m}^3$, $\mu = 1.85 \times 10^{-5} \text{ kg/ms}$) flows over a 2.0 m wide plate at 10.0 m/s velocity. Determine (i) shear stress (τ_0) and boundary layer thickness (δ) at a place where the boundary layer ceases to be laminar (ii) force on one side of the plate in the laminar region (iii) boundary layer thickness at 10 m from the leading edge. (6)